

Project Bill of Materials (BOM) - 270 Bolla Ave

Project Name: 270 Bolla Ave
Document Type: Integrated Systems Specification
Integrator Draft: v1.5 (NUC Standardization & Controller Roles)

Revision History

Version	Date	Description	Status
v1.0	2026-01-26	Initial Draft (Core Infrastructure)	Archived
v1.1	2026-01-27	Panel Consolidation & Door Entry Re-Spec	Archived
v1.2	2026-01-27	Unified DC Security Island & Environmental Hub	Archived
v1.3	2026-01-27	Official Part Verification & Manufacturer Hyperlinks	Archived
v1.4	2026-01-27	Final Systematic URL Verification & Project Correction	Archived
v1.5	2026-01-27	ASUS NUC 13 Standardization across LCP-2/LCP-3	Current

1. Project Objectives & System Organization

The automation system at 270 Bolla Ave is designed for maximum reliability, security hardening, and high-fidelity environmental control. The architecture is organized into three localized **Load Control Panels (LCPs)** to minimize home-run copper and isolate failure domains:

- LCP-1 (Garage):** The "Power Hub" for central living areas, kitchen lighting, and high-inrush holiday circuits.
- LCP-2 (Office 2):** The "Environmental Hub" and Master Wing controller. Houses weather station logic, PM2.5 wildfire interlocks, and master bedroom lighting.
- LCP-3 (Tech Room):** The "Security Island." A 100% battery-backed, DC-powered enclosure that manages 3D face recognition and entry without cloud dependency.

Core Standards: All lighting is **DALI-2** (addressable). All logic and switching is **KNX** (distributed). Infrastructure is "Future-Proofed" via stealth pre-wiring for window shades and seasonal exterior lighting.

Section 1: High-Voltage (HV) Infrastructure

Main Service & Smart Panel Telemetry

External MPN	Manufacturer	Internal ID	Description	Qty	Location	Notes
1-03294-02	SPAN	SPAN-GEN2	Smart Panel Gen 2 (32-Circuit)	2	Garage	Main Service (Side-by-side)
1-03294-02	SPAN	SPAN-SUB	Smart Panel Gen 2 (32-Circuit)	1	North Wall	Sub-Panel for Left Wing

13103701	Southwire	SER-1-AL	#1 AWG Aluminum SER Cable	-	Field	Sub-SPAN Feed from Main
28828201	Southwire	NM-B-12/2	12/2 Romex (High Current Loads)	-	Field	Direct to SPAN (Fridge, Range, etc)

Section 2: Load Control Panel 1 (LCP-1) (Garage) - Kitchen/Main Hub

The automation core for the central living areas.

- **Enclosure:** Saginaw SCE-24H2408LP
- **Dimensions (Outside):** 24.00" x 24.00" x 8.00" (610 x 610 x 203 mm)
- **Dimensions (Inside Usable):** 24.00" x 24.00" x 7.75" (610 x 610 x 197 mm)
- **Depth Note:** 8" (203mm) nominal depth easily accommodates the 128.5mm (5.1") Mean Well SDR-480.

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
SCE-24H2408LP	Saginaw	SAG-24H2408	24x24x8 NEMA 4/12 Enclosure	1	Hinged Cover
SDR-480-24	Mean Well	MW-SDR-480	480W 24V DC PSU (UL 508)	1	Main Logic Power
SV/S 30.640.5.1	ABB	ABB-KNX-PS	KNX Power Supply 640mA	1	Powers Line 1.1
LK/S 4.2	ABB	ABB-LK-S	KNX Line Coupler	1	Connects 1.1 to Backbone
CBM E4 24DC/0.5-10A-NO	Phoenix Contact	PHX-CBM-E4	Electronic Circuit Protector	1	NEC Class 2 Compliance
5WG1141-1AB03	Siemens	SIE-DALI-GW	KNX/DALI Gateway Twin N 141/03	1	Universes 1 & 2
1060/A	eldoLED	ELD-PWR-100	100W DALI-2 LED Driver	6	Garage/Kitchen/Living Tape
86458668	Lunatone	LUN-DALI-010	DALI 0-10V PWM Interface	1	Powder Room Fan Control

Section 3: Load Control Panel 2 (LCP-2) (Office 2) - Master Wing Hub

The automation core for the Master Bedroom, Bedrooms 3/4, and local environmental logic.

- **Enclosures:** nVent HOFFMAN ASE24X12X4NK / ASE12X12X4

- **Inside Dimensions:**
 - Top (24x12): 23.8" x 11.8" x 3.87" (605 x 300 x 98 mm)
 - Bottom (12x12): 11.8" x 11.8" x 3.87" (300 x 300 x 98 mm)
- **Depth Note: MAX COMPONENT DEPTH 3.5" (89mm).** No SDR-series PSUs allowed here; use MDR or SV/S series only. (NUC fits @ 37mm).

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
ASE24X12X4NK	nVent HOFFMAN	HOF- ASE24X12X4	Automation Enclosure (24x12x4)	1	LCP-2 Top Box (DIN Rail)
ASE12X12X4	nVent HOFFMAN	HOF- ASE12X12X4	Network/Aux Enclosure (12x12x4)	1	LCP-2 Bottom Box (Switches)
NUC13ANKi5	ASUS	ASUS-NUC- 13	NUC 13 Pro (i5, 16GB RAM)	1	Master Logic Controller (KNX/DALI Hub / HomeWorks-Class Automation)
SV/S 30.640.5.1	ABB	ABB-KNX-PS	KNX Power Supply 640mA	1	Powers Line 1.2
LK/S 4.2	ABB	ABB-LK-S	KNX Line Coupler	1	Connects 1.2 to Backbone
5WG1141- 1AB03	Siemens	SIE-DALI-GW	KNX/DALI Gateway Twin	1	Universes 3 & 4
1060/A	eldoLED	ELD-PWR- 100	100W DALI-2 LED Driver	6	Master Bed Accent/Coves
86458668	Lunatone	LUN-DALI- 010	DALI 0-10V PWM Interface	4	Bath/Laundry Fan Controls

Section 3.1: Load Control Panel 3 (LCP-3) (Tech Room) - Door Entry MDF

The centralized "Security Island" - 100% DC-backed for maximum uptime.

- **Enclosure Option A (Standard):** Saginaw SCE-24H2408LP (8" / 203mm Depth)
- **Enclosure Option B (Flush-Clean):** Saginaw SCE-24H2406LP (6" / 152mm Depth)
- **Dimensions (Inside Usable A):** 24.00" x 24.00" x 7.75" (610 x 610 x 197 mm)
- **Dimensions (Inside Usable B):** 24.00" x 24.00" x 5.75" (610 x 610 x 146 mm)
- **Depth Rationale:**
 - **Bottleneck 1:** TP-Link Industrial Switch @ 120mm (4.7") depth.
 - **Bottleneck 2:** Yuasa 12Ah Batteries @ 98mm (3.86") depth.
 - **Wall Context:** A 2x6 wall (5.5" stud) + drywall (~6" total) makes Option B (6") nearly flush.

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
SCE-24H2408LP	Saginaw	SAG-24H2408	24x24x8 NEMA Enclosure	1	Hinged Cover (Security MDF)
eFlow104N	Altronix	ALT-EFLOW104N	10A Power Supply/Charger	1	Managed DC Rail
NP12-12	Yuasa	BATT-12AH	12V 12Ah SLA Battery	2	24V String (Powers NUC/Switch/Locks)
DDR-60G-15	Mean Well	MW-DDR-60G-15	24V to 15V DC-DC Converter	1	Power for NUC (Adj. to 18V)
NUC13ANKi5	ASUS	ASUS-NUC-13	NUC 13 Pro (i5, 16GB RAM)	1	Security Brain (Akuvox SDMC Server / 3D Door Entry)
IES210GPP	TP-Link	TPL-IES210GPP	Industrial 10-Port PoE+ Switch	1	DC-Input (12-55V) PoE Source
SR01	Akuvox	AKU-SR01	Secure Relay Module	4	DIN-Rail Mounted via Clips
BE/S 16.20.3.2	ABB	ABB-BIN-16	16-fold Binary Input	1	Security/Status Monitoring

Section 4: KNX Infrastructure (Field)

User Interfaces and Sensors

4.1 Backbone & IP Connectivity

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
IPS/S 3.1.1	ABB	ABB-IPS-S	KNX IP Interface Secure	1	In LCP-2 - High-Speed Logic Link
USB/S 1.2	ABB	ABB-USB-S	KNX USB Interface	1	In LCP-2 - Local Diagnostic Backup
SV/S 30.160.1.1	ABB	ABB-PS-160	KNX Power Supply 160mA	1	Powers the Main Backbone Line

4.2 Sensors (Environmental & Presence)

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
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056353	Steinel	STE-TP-KNX	True Presence Multisensor KNX	4	Primary Bath, Guest, Bath 3, Laundry
BE/S 16.20.3.2	ABB	ABB-BIN-16	16-fold Binary Input	2	Door/Window Contacts (In LCPs)

4.3 User Interfaces & Cabling

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
Sentido 4-way	Basalte	BAS-SENT-4	Sentido (4-Button) Switch	12	Minimal code-required placement
KNX Bus Coupler	Basalte	BAS-BUS-COU	KNX Bus Coupler for Sentido	12	Required for switch connection
UNITRONIC BUS EIB/KNX	Unitronic	CAB-KNX-G	KNX Green Bus Cable (UL)	1	1000ft Spool
243-211	Wago	WAG-KNX-CONN	KNX Push-Wire Connectors	2	Red/Black (Boxes of 50)

Section 5: Network & AI Compute (Office 2 / MDF Hub)

Secondary compute and data storage in rack.

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
NUC13ANKi7	ASUS	ASUS-NUC-AI	NUC 13 Pro (i7, 32GB RAM)	1	AI Compute Node (Pattern Learning/LLM)
MZ-V9P4T0B-AM	Samsung	SSD-990-4TB	990 Pro 4TB NVMe SSD	1	High-endurance for AI/Logging
USW-Pro-Max-24-PoE	Ubiquiti	UI-SW-PRO-24	UniFi Pro Max 24 PoE	1	Core PoE Backbone
UVC-G5-Pro	Ubiquiti	UI-CAM-G5P	G5 Professional 4K Camera	12	12x Camera home-runs required
PDU15B10R	CyberPower	CYB-PDU-1U	1U Rack PDU	1	Powers Server & Network Gear
SMT1500C	APC	APC-UPS-1500	Smart-UPS 1500VA	1	Backup for Linux Box & AI Nodes

Section 6: Ventilation & Fans

Whole-House, Attic, and Bathroom Exhaust (EC Motors + DALI-2 Control)

6.1 Whole-House Fans (WHF)

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
Stealth Pro 7.0X	QuietCool	QC-STL-7.0X	Stealth Pro 7.0X (ECM Motor)	2	1 per arm of U-shape
SR-2701S-DT7	Sunricher	SUN-2701S-DT7	DALI-2 Relay Puck	2	UR Recognized - Power On/Off
SR-2303-0-10V-PWM	Sunricher	SUN-2401-10V	DALI to 0-10V Signal Converter	2	UR Recognized - Variable Speed

6.2 Attic Exhaust Fans

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
AFG SMT PRO-3.0	QuietCool	QC-AFG-3.0	AFG SMT PRO-3.0 Gable Fan	4	2 per gable end
SR-2701S-DT7	Sunricher	SUN-2701S-DT7	DALI-2 Relay Puck	4	UR Recognized - Power On/Off
SR-2303-0-10V-PWM	Sunricher	SUN-2401-10V	DALI to 0-10V Signal Converter	4	UR Recognized - Variable Speed

6.3 Bathroom & Laundry Exhaust

Location	Fan Motor	Silencer	Backdraft Damper	Diffuser (InviAir)
Primary Bath	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (60") + S100 (6")
Bath 2 (Kids)	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (48")
Bath 3 (Ens)	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (48")
Guest Bath	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (60")
Laundry Room	Fantech FG 6M EC	Fantech LD 6	RSK 6	S100 (8")
Half Bath	Fantech FG 6M EC	Fantech LD 6	RSK 6	S100 (6")

6.4 Environmental AI Inputs

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
056353	Steinel	STE-TP-KNX	True Presence Multisensor KNX	4	Primary, Guest, Bath 3, Laundry
2064965	Warema	WAR-WETH-PRO	KNX Weather Station Pro	1	Outdoor (North Wall)
OAL-PM2.5	Temco	TEM-PM25	Outdoor PM2.5 Sensor (Modbus)	1	Wildfire Smoke Interlock
SCN-MBGW.01	MDT	MDT-SCN-MB	KNX/Modbus Gateway	1	Bridge for PM2.5 Data

Section 7: Field Devices - Lighting (DALI-2)

Based on current floor plan placements.

External MPN	Manufacturer	Internal ID	Description	Qty	Type	Notes
X-Series	DMF Lighting	DMF-X2-SQ-FL	X-Series Square Flangeless	80	Recessed	UL Listed (IC-rated/Wet)
X-Series Wet	DMF Lighting	DMF-X2-WET	X-Series Sq Flangeless (Wet)	4	Recessed	1 per Shower/Tub
Haiku 52"	Big Ass Fans	BAF-HAIKU-52	Haiku 52" Aluminum	3	Fan+Light	Requires 0-10V Int
PCS-DUO	Southwire	CAB-NM-PCS	Romex PCS Duo (14/2 + 16/2)	3	Spools	Power+Control for Lights & Fans

Section 8: Bulk Infrastructure & Consumables

Item	Manufacturer	Description	Qty	Notes
CAB-KNX-G	Unitronic	KNX Green Bus Cable (UL)	TBD	CMG/PLTC Rated
CAB-CAT6-RIS	TrueCable	Cat6 Riser (Bulk)	TBD	UL Listed (E497331)
CAB-16/4-SHLD	Southwire	16/4 Shielded (Shade Pre-wire)	TBD	Home-run to LCP-1/2
BOX-SQ-4IN	Hubbell	4" Square Steel Box (2-1/8" D)	-	Deep boxes for DALI pucks

BOX-FIRE-PAD	STI	SpecSeal Putty Pad	1	Required for Garage Intercom
CON-EMT-1IN	Western Tube	1" EMT Conduit	2	Solar Roof-Runs
CON-EMT-2.5IN	Western Tube	2.5" EMT Conduit	2	Energy Wall Super-Conduit

Section 9: Door Entry & Access Control (Akuvox)

3D Face Recognition & SIP Intercom System

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
X915S	Akuvox	AKU-X915S	8" 3D Face Recognition Intercom	1	Front Door (CE/FCC)
E16C	Akuvox	AKU-E16C	5" 3D Face Recognition Intercom	3	Garage/Side/Laundry
E16-A05H	Akuvox	AKU-E16-HOOD	Sun Shield / Rain Hood	1	For Garage Man Door
S567G	Akuvox	AKU-S567G	10" Android 12 Indoor Monitor	1	Hallway Tablet (Main)
1006-CS	HES	HES-1006	Heavy Duty Electric Strike	4	UL 1034 Listed
SR01	Akuvox	AKU-SR01	Secure Relay Module	4	Centralized in LCP-3

Section 10: Motorized Skylights & HVAC Integration

Centralized 24V DC Actuation & Weather Logic

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
JAL-0810.02	MDT	MDT-JAL-0810	8-fold Shutter Actuator (24V DC)	1	KNX Certified (No UL)
INKNXMBM100	Intesis	INT-MOD-KNX	HVAC Modbus to KNX Gateway	1	Bridge for AC/Heat
MDR-100-24	Mean Well	MW-MDR-100-24	100W 24V DC DIN-Rail PSU	1	UL 508 Listed
2064965	Warema	WAR-WETH-PRO	KNX Weather Station Pro	1	Home-runs to LCP-2

OAL-PM2.5	Temco	TEM-PM25	Outdoor PM2.5 Sensor (Modbus)	1	Home-runs to LCP-2
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Section 11: Routing & Pro-Integrator Notes

11.1 KNX Bus Topology (Redundant Loop Strategy)

- **Perimeter Run:** Route KNX Green Bus through all habitable walls at **48" AFF** (Switch Height).
- **Ceiling/Attic Spine:** Provide a central bus run through the attic with drops to all bathroom sensor locations.
- **Hardened Loop:** All major bus segments must be pulled as a **"Terminated Loop"**. Run the cable from the LCP, through the zone, and back to the LCP. **CRITICAL:** Only one end is to be connected to the bus power supply; the return tail must be labeled and capped.

11.2 LCP Service Spares & Pre-Wire

- **Window Shades (Pre-Wire):** Pull **16/4 Shielded Cable** from LCP-1/2 to every window header. Terminate in "Router & Rescue" vaults per *window-shade-wiring-prep.md*. Coil 3ft of slack in LCP, label, and do not terminate.
- **LCP-1 (Garage):** Pull **3x Spare Service Loops** (120V HV In + DUO Control Out) into the attic.
- **LCP-2 (Office 2):** Pull **3x Spare Service Loops** (120V HV In + DUO Control Out) into the attic.
Environmental Hub: Weather station and PM2.5 sensors terminate here.

11.3 Box & Finish Standards

- **Deep Boxes:** Standardize on 2-1/8" deep steel boxes to allow for DALI relay pucks and KNX pigtail service loops.
- **Floating Shields:** Drain wires on the KNX bus should be cut and taped at the device end; bond only at the LCP ground rail to prevent ground loops.

11.4 Security Puck (SR01) Placement

- **Anti-Tamper Protocol:** All **Akuvox SR01** security relays must be centralized in **LCP-3** (Tech Room).
- **NO LOCAL LOGIC:** Under no circumstances should the relay be placed near the door terminal. The connection from door to panel is RS485 encrypted data only. This prevents "paperclip" attacks on the lock hardware.
- **Mounting:** All pucks to be DIN-rail mounted using official clips. Label each with the corresponding door name.

Section 12: Patio Infrastructure (Pre-Wire Only)

High-Current Heating & Outdoor Fan Control

External MPN	Manufacturer	Internal ID	Description	Qty	Notes
28829001	Southwire	CAB-10/2-NM	10/2 Romex (30A 240V)	3	Sub-SPAN to Heater J-Box
PCS-DUO	Southwire	CAB-NM-PCS	Romex PCS Duo (14/2 + 16/2)	2	LCP-2 to Patio Fans

18/2 SHLD	Southwire	CAB-DALI-G	18/2 Shielded (DALI Bus)	1	LCP-2 to Heater J-Box
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12.1 Patio Implementation Notes

- **Heaters (18kW Plan):** Pull 3x 30A dedicated circuits. Control is via DALI-2 relay/contactor logic (hardware deferred).
- **Fans:** Pull power + control (Duo) for 2x Outdoor Haiku fans.
- **Surge Protection:** All outdoor data/bus lines must include a DIN-rail surge protector at the LCP entry point.

Section 13: Known Missing & Future-Phase TBDs

13.1 Aesthetic & Decorative Lighting (Pending Selection)

- **Chandeliers / Pendants:** Dining, Entry, and Kitchen Island fixtures are TBD. **Requirement:** All decorative junction boxes must be 2-1/8" deep to accommodate DALI-2 relay/dimmer pucks.
- **Vanity Sconces:** All bathroom vanity lighting locations and fixtures are TBD.
- **Outdoor Architectural Sconces:** Front entry and perimeter wall sconces are TBD.

13.2 Seasonal & Holiday Lighting (KNX Controlled)

- **Eave Outlets:** 2x Weatherproof outlets mounted under the roof eaves (North/South peaks) for Christmas/Holiday lighting. Home-run to **LCP-1**.
- **Ground Seasonal Outlets:** 2x Weatherproof outlets at ground level (Front/Side) for seasonal displays. Home-run to **LCP-1**.
- **Control Logic:** These 4 outlets must be terminated on a **KNX High-Inrush Relay Actuator** in LCP-1 for automated seasonal scheduling.

13.3 Exterior Infrastructure (Future Wiring)

- **Landscape Spine:** Pull 1" EMT conduit from Garage (LCP-1) to an external NEMA-3R landscape junction box for future landscape lighting expansion.
- **Under-Awning Future-Proofing:** Pull **14/2 Southwire Duo** to 4 corners of the exterior roofline for future architectural accent lighting.
- **Security Camera Drops:** Ensure all 12 Ubiquiti camera locations (Section 5) have Cat6 home-runs to the Tech Room (LCP-3).

13.4 Closet & Storage Automation (WIC / Linen / Pantry)

- **Reed Switches (Primary):** Install recessed magnetic contact switches (e.g., Tane Pill-V) in all closet door frames. Home-run 18/2 to the nearest LCP Binary Input.
- **Stealth mmWave Pre-Wire:** Pull a KNX Bus "tail" to the center ceiling of each WIC and Linen closet. Coil 2ft of slack **above the drywall** (Stealth Vault).
- **Future sensor:** Prepared for **Creatrol 24G/60G mmWave** hidden installation (drywall penetration mode).
- **Requirement:** NO physical light switches are to be installed in closets or small storage areas. Lighting must be 100% sensor/contact driven.

Section 14: Products Struggling / TBD Direct Links

The following products have generic or reseller links and require further direct manufacturer verification:

- **Akuvox E16-A05H Hood:** Currently linked to Low Voltage Dealer. Direct Akuvox accessory page not found.
- **Unitronic KNX Green Bus:** Current link is LAPP region-specific. Need a global spec sheet for North American compliance (LAPP USA).
- **Warema Weather Station Pro:** Product ID 2064965 confirmed, but technical documentation is region-locked.
- **MDT SCN-MBGW.01:** Verified as correct Modbus Gateway; need to confirm specific firmware features for AI PM2.5 interlock.

Section 15: Deferred Lighting & Aesthetic Specification (TBD)

Items to be finalized in Phase 1B / Interior Finish Phase.

Area	Lighting Type	Control	Status	Notes
Main Garage	High-Lumen LED Tape	DALI-2 (via LCP-1)	Deferred	Driven by SDR-480-24 in LCP-1.
Dining / Entry	Decorative Pendants	DALI-2 Relay/Puck	TBD	Requires 2-1/8" deep J-Boxes.
Bath Vanities	Task Sconces	DALI-2 Relay/Puck	TBD	Fixture selection pending.
Outdoor Walls	Architectural Sconces	DALI-2 Relay/Puck	TBD	Aesthetic match for Akuvox entry stations.